L-4742 47314

UNIT L-53147
S.O. 82231

CB-200, 15 lb Steam

# FORM H-2 MANUFACTURERS' DATA REPORT FOR ALL TYPES OF BOILERS EXCEPT WATER-TUBE AND THOSE MADE OF CAST IRON

## As Required by the Provisions of the ASME Code Rule

C10-0004A 2/71

Part No. 270-435

1. Manufactured by Cleaver-Brooks Division of Aqua-Chem, Inc., Milwaukee, Wisconsin
(Name and address of manufacturer)

2. Manufactured for Passaic Valley Water Commission Little Falls, New Jersey
(Name and address of purchaser)

3. Type Internally Fired (H.R.T.-Loco.-Compact-Marine, Miniature) Boiler No. 47514 (Mfrs. Serial No.) (State and State No.) 25507 (Natl. Board No.) Year Built 1971

4. The chemical and physical properties of all parts meet the requirements of material specifications of the ASME BOILER AND PRESSURE VESSEL CODE. The design, construction, and workmanship conform to ASME Rules, Section IV Dated 19 71

Remarks: Manufacturers' Data Records properly identified and signed by Commissioned Inspectors have been furnished for the following items of this report:
(Name of Part — Item number, manufacturer's name, and identifying stamp)

We certify the statement in this data report to be correct.

Date September 16, 19 71 Signed Cleaver-Brooks Division (Manufacturer) By Earl R. Stoudt
Ear A. Stoudt

Certificate of Authorization Expires XXXXXXXXX 19
1-15-74

## CERTIFICATE OF SHOP INSPECTION

BOILER MADE BY Cleaver-Brooks Division at Lebanon, Pennsylvania

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and/or the State or Province of ............ and employed by Hartford Steam Inspection & Insurance Co. of Hartford, Conn. have inspected parts of this boiler referred to as data items 1-9, 11, 16, & 19-23 and have examined manufacturer's partial data reports for items None and state that, to the best of my knowledge and belief, the manufacturer has constructed this boiler in accordance with the applicable sections of the ASME BOILER AND PRESSURE VESSEL CODE.

By signing this certificate neither the Inspector nor his employer makes any warranty, expressed or implied, concerning the boiler described in this manufacturer's data report. Furthermore, neither the inspector nor his employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date September 16, 19 71

Frank Bergholm
Frank Bergh[illegible]

Commissions PA WC 560 NB 3113
Nat'l Board or State and No.

We certify that the field assembly of all parts of this boiler conforms with the requirements of SECTION IV of the ASME BOILER AND PRESSURE VESSEL CODE.

Date .............. 19 ...... Signed (Assembler) By (Representative)

Our Certificate of Authorization to use the (A) or (B) Symbol Expires 19

## CERTIFICATE OF FIELD ASSEMBLY INSPECTION

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and/or the State or Province of ............ and employed by ............ of ............ have compared the statements in this manufacturer's data report with the described boiler and state that the parts referred to as data items ............, not included in the certificate of shop inspection have been inspected by me and that to the best of my knowledge and belief the manufacturer and/or the assembler has constructed and assembled this boiler in accordance with the applicable sections of the ASME BOILER AND PRESSURE VESSEL CODE. The described boiler was inspected and subjected to a hydrostatic test of ............ psi.

By signing this certificate neither the Inspector nor his employer makes any warranty, expressed or implied, concerning the boiler described in this manufacturer's data report. Furthermore, neither the inspector nor his employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date .............. 19 ......

Inspector Commissions ............ Nat'l Board or State and No.

5. Boiler Shells or Drums No. 1 Diam. 60" Length 168 in. Diam. ............ Length ............ ft. ............ in.

6. Shell Plates SA[illegible] 5/16" (For each Drum, state: Brand Manufacturer's Brand, Material Specification Number, Thickness)

7. (a) Longitudinal Joint Fusion double welded butt (Seamless, Fusion or Forge Welded, Riveted-Lap or Butt-Single, Double, Triple or Quadruple)

(b) Rivets ............ (Diameter and Material Specification No.) Rivet Holes ............ (Diameter and Pitch) Eff. of Long. Joint 85 % (As compared to shell plate)

8. (a) Tube Sheet SA-36, 1/2" (Brand, Mat. Spec. No.; Thickness) Reinforcement ............ (No. and Thickness of Straps) Rivets Fillet weld (No. Rows; Mat. Spec. No.; Hole Diam. & Pitch)

(b) Tube Holes 2.520" (Diameter) Pitch ............ (Longitudinal and Girth) Eff. of Tube Ligament ............ % (Compared to shell plate)

9. Girth Joints Fusion double welded butt (Seamless, Fusion or Forge Welded, Riveted-Lap or Butt) Rivet Holes ............ No. of Courses 2

10. Heads ............

11. Boiler Tubes: No. SA-178-A, 108, Straight (Mat. Spec. No.-Straight or Bent) Diam. 2-1/2" Length 168-1/2" (If various, give maximum and minimum lengths) Gage .105" (or thickness)

12. Headers: No. ............ Heads or Ends ............ Hydro. Test ............ lb.

13. Mud Drum ............ Heads or Ends ............ Hydro. Test ............ lb.

14. (a) Waterwall: Headers, No. ............ Heads or Ends ............ Hydro. Test ............ lb.

(b) Tube Holes ............ Tubes ............ Gage ............ Heating Surface ............ sq. ft.

15. (a) Superheater: Headers, No. ............ Heads or Ends ............ Hydro. Test ............ lb.

(b) Tube Holes ............ Tubes ............ Gage ............

16.

| Stays or Braces | Material Spec. No. | Type | No. | Size | Welded or Weldless | Total Net Area | Fig. P-28 L/1 | Dist. Tubes to Shell | Area to be Stayed | Max. S.W.P. |
|---|---|---|---|---|---|---|---|---|---|---|
| (a) F.H. above tubes | | | 0 | | | 0 | | 16-1/2" | 0 | 15 lb |
| (b) R.H. above tubes | | | 0 | | | 0 | | 16-1/2" | 0 | 15 lb |
| (c) F.H. below tubes | | | | | | | | | | |
| (d) R.H. below tubes | | | | | | | | | | |
| (e) Through below tubes | | | | | | | | | | |
| (f) Dome Braces | | | | | | | | | | |

17. Staybolts ............ Pitch ............ Max. S.W.P. ............

18. (a) Dome: Diam. ............ Longitudinal Seam ............ Eff. ............ % How secured to shell ............

(b) Opening in shell ............ Head ............

19. (a) Furnace No. 1 Size 24" O.D. (Int. Diam. or LxWxH) Length, each Section, ............ in., ............ in., Total 168¼" Type Plain (Plain, Adamson, or Corr.)

SA-53B, .312" (Brand, Mat. Spec. No., Thickness) (b) Seams: Type Seamless (Seamless Fusion or Forge W., Riveted, Lap, or Butt) Location ............ (Seam)

20. Openings: (a) Steam 1 @ 10" 150 lb Flange (No., Size, and Type of Nozzles or Outlets) (b) Safety Valve 2 @ 2-1/2" (No., Size, and Type of Nozzles or Outlets)

(c) Blowoff 2 @ 2" NPT bottom ¢ (No., Size, Type and Location of Outlets) (d) Feed 2 @ 2" NPT (No., Size, Type, and Location of Connections)

(e) Manholes: No. 1 Size 11" x 15" Location Shell – Ring (How Reinforced)

(f) Handholes: No. 6 Size 3-1/4" x 4-1/2" Location Shell (How Reinforced)

21. Fusible Plug (If Used) ............ (No., Diam., Location, Mfgr. Stamp) Boiler Supports: No. 4 Legs – Welded (Hangers or Lugs; Riveted or Welded)

22. Max. AWP 15 psi Based on HG-312 (Code Par. and/or Formula) Heating Surface 1000 sq. ft. (Total)

23. Hydrostatic Test: (a) ............ lb. (Welded Drums) (b) ............ lb. (Riveted Drums) (c) 60 lb. (Completed Boiler)